

CS5803 Projects

Sl. No.	Title	Description	References	Contact
1	Simplifying Text	Document-level text simplification is a specific type of simplification that involves simplifying documents consisting of several sentences by rewriting them into fewer or more sentences. In this project, we will aim to simplify an input text which might be difficult to comprehend. Such texts often occur in legal documents, domain specific documents, etc.	Dataset: https://github.com/epfml/easy-summary/tree/main/SimSum/data Paper: SIMSUM: Document-level Text Simplification via Simultaneous Summarization	Prof. Maunendra/Maharaj maunendra@cse.iith.ac.in s23resch01004@iith.ac.in
2	Detecting hallucination	Natural sequence models are known to hallucinate. The objective of the work is to detect hallucinations in natural language generation models.	Papers & Dataset: https://arxiv.org/pdf/2402.10496.pdf https://arxiv.org/pdf/2104.08704.pdf Blog: https://www.amazon.science/blog/new-tool-dataset-help-detect-hallucinations-in-large-language-models	Prof. Maunendra/Maharaj maunendra@cse.iith.ac.in s23resch01004@iith.ac.in
3	Bilingual dictionaries plus LLM for low-resource machine translation	Large language models are shown to have the capability to perform language translation without being trained on parallel data. However, this capability is limited for under-represented languages, called low-resource languages. One of the	Dataset: https://github.com/openlanguage/data/flores Paper: Dictionary-based Phrase-level Prompting of Large Language	Maharaj cs23resch01004@iith.ac.in

		potential approaches to improve translation is to utilize less-costlier bilingual dictionaries along with a prompt to guide cross-lingual transfer between languages. In this work, the first objective is to build bilingual dictionaries (including domain-specific) for truly low-resource languages using existing methods. The second objective is to perform comprehensive experiments with various closed and open-source LLMs. The languages should be considered from the FLORES-200 dataset.	Models for Machine Translation	
4	Rumor Verification using Evidence from Authorities	Given a rumor expressed in a tweet and a set of authorities (one or more authority Twitter accounts) for that rumor, represented by a list of tweets from their timelines during the period surrounding the rumor, the system should retrieve up to 5 evidence tweets from those timelines, and determine if the rumor is supported (true), refuted (false), or unverifiable (in case not enough evidence to verify it exists in the given tweets) according to the evidence. This task is offered in both Arabic and English. (Note: This is an ongoing Shared Task at CLEF 2024)	Dataset: https://checkthat.gitlab.io/clef2024/task5/ Shared task: https://checkthat.gitlab.io/clef2024/task5/	Shrey ai22mtech02003@iith.ac.in
5	News Image Caption Generation	News-image captioning is the task of generating a description of an image given the image and its article body as	Dataset: https://github.com/furkanbiten/GoodNews	Debolena ai20resch11003@iith.ac.in

		input. This task is more challenging than conventional image captioning because it requires a joint understanding of image and text.	Paper: Image Caption Generation for News Articles Knowledge Prompt Makes Composed Pre-Trained Models Zero-Shot News Captioner Image Caption Generation for News Articles GoodNews	
6	Tone Identification in written text for Bodo language	Bodo is a tonal language with two tones: High and Low. However, unlike other tonal languages, this linguistic characteristic is not reflected in the standard written form. These tonal markers can play an especially important role in tasks such as text-to-speech, L2 language learning, and word sense disambiguation. The main objective of the project is, given a Bodo sentence, to classify whether the tone of each tonal word is High or Low. (Note: This is a less explored problem for the Bodo language and can be considered a novel task. However, prior knowledge of the Bodo language itself is not required to carry out the project. The project might also require the creation of novel lexical resources.)	Dataset: https://drive.google.com/file/d/1QR3Z3MpDGW8xS154IUE9otXh7ckiHmW/view?usp=sharing Paper: Some ascepts of the Tonal Phonology of Bodo	Maharaj cs23resch01004@iith.ac.in

7	Detecting toxicity in recent open-sourced LLMs (not ChatGPT/Gemini)	The objective of this work is to detect toxicity in open-sourced large language models.	Dataset: Real Toxicity Prompts Paper: Real Toxicity Prompts: Evaluating Neural Toxic Degeneration in Language Models	Debolena ai20resch11003@iith.ac.in
8	Detecting Toxic content from social media using LLMs	The objective of this work is to detect toxic social media content by utilizing large language models.	Papers & Dataset: Efficient Toxic Content Detection by Bootstrapping and Distilling Large Language Models Toxicity Detection with Generative Prompt-based Inference	Debolena ai20resch11003@iith.ac.in
9	Indic Language Identification: Benchmarking and Improving	Language Identification (LID) is a critical task in Natural Language Processing, serving as a prerequisite step for all multilingual NLP tasks. LID presents particular challenges for low-resource languages, especially when they share the same writing script. Existing models, like IndicLID, have shown limitations in accurately distinguishing between Hindi and Bodo sentences. This work aims to: 1. Benchmark various LID tools/models. 2. Develop algorithms specifically designed to improve low-resource Indian language identification, with a particular	Dataset: https://github.com/AI4Bharat/IndicLID/tree/master Papers: Bhasha-Abhijnaanam: Native-script and Romanized Language Identification for 22 Indic Languages An Open Dataset and Model for Language Identification	Maharaj cs23resch01004@iith.ac.in

		focus on languages sharing similar scripts.		
10		Data Contamination is a situation where the test set of a downstream task is present in the training data of LLMs. This is a major issue in measuring the effectiveness of LLMs' performance. In recent work, most of the research has focused on detecting data contamination for English. However, detecting data contamination for other languages in a Multilingual setting is imperative. The work aims to explore various LLMs and detect data contamination for languages other than English.	Dataset: Standard benchmark dataset should be selected as per the task considered.	
	Data Contamination in Large Language Models: Finding it beyond English	(Note: This project is expected to be comprehensive in terms of languages and LLMs considered)	Paper: Time Travel in LLMs: Tracing Data Contamination in Large Language Models	Maharaj cs23resch01004@iith.ac.in

11	Multistep reasoning over text (or knowledge base)	Reasoning using knowledge is a common task for humans. However state of the art language models struggle to reason over knowledge. For example, language models are poor at math. Generating a step by step reasoning for reaching a conclusion also addresses the interpretability problem of LLMs. Effective multi-step reasoning can improve trust worthiness of LLMs while the ability to reason over very large number of steps can give new capabilities to LLMs like theorem proving or policy making.	Dataset: Entailment bank Paper: Explaining Answers with Entailment Trees	Dishank ai20btech11011@iith.ac.in
12	End To End Product Retrieval	1a) End to end pipeline with synthetic query generation (through simulation, possibly using LLM) 1b) Retrieve relevant documents(products) for the query (prefer higher relevance label) <i>Other related tasks to help with main task</i> 1) Classification task to identify if product is relevant to query 2) Given query recommend products in decreasing order of relevance 3) Generating relevant sythetic queries for products	Train,val data with query id, document id (product), relevance label 1) Queries(~31k) with associated query id (qids) 2) Collections dataset (~1.7M) - product/documents and descriptions with associated document ids 3) Triplets dataset (30k, 6Gb) - query, positive items and negative items Project page	Aishwarya ai21resch11002@iith.ac.in

13	Clickbait spoiler detection	Clickbaits are texts containing sensationalized terms that aim at raising the curiosity of the readers, and lure them to click on a hyperlink. These are common in news headlines, social media posts etc. This project will target identifying clickbaits and generating spoilers for the same.	Webis Clickbait Spoiling Corpus 2022 Generating clickbait spoilers with an ensemble of large language models TohokuNLP at SemEval-2023 Task 5: Clickbait Spoiling via Simple Seq2seq Generation and Ensembling	Prof. Maunendra maunendra@cse.iith.ac.in
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